

Transcription

Current 46-page faculty deck primary - official Week 4 objectives integrated - prokaryotic/eukaryotic transcription and RNA processing tied to drug targets and disease

4 official objectives

Sigma & promoters

Pol I/II/III

Rifampin

Alpha-amanitin

mRNA processing

SOURCE PRIORITY: the uploaded current faculty deck controls terminology, examples, and emphasis. The official Week 4 learning-objective sheet defines what must be mastered. This companion reorganizes the same material for retrieval and application without replacing the source deck.

Official Week 4 Objectives

1. Summarize initiation, elongation, and termination of transcription and compare prokaryotic with eukaryotic transcription.
2. Identify the molecular targets of rifampin and alpha-amanitin.
3. Describe post-transcriptional processing of eukaryotic mRNA and explain how defects in processing can cause disease.
4. Identify transcription/RNA-processing defects relevant to systemic lupus erythematosus and autosomal-dominant retinitis pigmentosa and connect them to therapeutic thinking.

ASSIGNED SCHOLARRX

DNA Transcription & Translation

- RNA polymerase synthesizes RNA **5' to 3'** using a DNA template.
- RNA synthesis begins **de novo** - no primer is required.
- The deck emphasizes that RNA polymerase lacks the 3' to 5' exonuclease proofreading system characteristic of replicative DNA polymerases.

Initiation

->

elongation

->

termination

->

eukaryotic RNA processing

Bacterial RNAP holoenzyme = core enzyme (2 alpha, beta, beta-prime, omega) + sigma factor. Sigma directs promoter recognition; beta is the catalytic subunit emphasized by the rifampin mechanism.

Typical bacterial promoters contain **-35** and **-10 (Pribnow box)** cis-regulatory elements recognized by the RNAP holoenzyme. Because bacteria lack a nucleus, transcription and translation are spatially and temporally coupled.

Termination mode	Core logic
Rho-independent (intrinsic)	RNA forms a terminator hairpin followed by a U-rich tract; the transcription complex destabilizes and RNA is released.
Rho-dependent	Rho helicase binds a rut site, uses ATP to move along RNA, catches the paused polymerase, and separates RNA from the DNA/template complex.

Polymerase	Main products emphasized in deck
RNA Pol I	45S pre-rRNA → 28S, 18S, 5.8S rRNA
RNA Pol II	pre-mRNA/mRNA, miRNA, many snRNAs
RNA Pol III	pre-tRNA/tRNA and 5S rRNA

Promoter / enhancer logic

Core promoter elements and transcription-factor binding sites recruit the transcription machinery. Enhancers can act at a distance by DNA looping; silencers/repressors decrease transcription.

Pol II initiation

TBP helps initiate assembly at the promoter. **TFIIH** has helicase activity that opens DNA and kinase activity that phosphorylates the Pol II CTD, promoting transition into elongation.

Eukaryotic compartment rule: transcription occurs in the nucleus while translation occurs in the cytoplasm, creating space and time for RNA processing before translation.

Slides 8 & 20

Drug and toxin targets

LO 2

Agent	Target	Effect
Rifampin / rifampicin	Bacterial RNAP beta subunit near active center	Blocks extension of very short nascent RNA; key antimycobacterial mechanism
Alpha-amanitin	Eukaryotic RNA Pol II (Pol III at higher dose)	Severe transcriptional inhibition; Amanita toxin with liver/kidney toxicity
Actinomycin D	DNA-dependent transcription broadly	Nonselective transcription inhibitor in deck summary

Exam pair: rifampin = bacterial beta subunit. Alpha-amanitin = eukaryotic Pol II.

Slides 23-25

5' m7G cap

Added by a capping complex associated with Pol II. Supports nuclear export, protects against degradation, and promotes translation.

Splicing

Introns removed and exons joined by the spliceosome.

3' poly(A) tail

Pre-mRNA is cleaved downstream of the AAUAAA signal and poly(A) polymerase adds the tail.

Nuclear export

Only appropriately processed transcripts are efficiently exported for cytoplasmic translation.

Cap functions: export, stability, translation. The deck notes that nuclear transcripts are capped; prokaryotic and mitochondrial mRNAs do not use this nuclear Pol II capping system.

The spliceosome is built from small nuclear ribonucleoproteins (snRNPs). Splice-site recognition depends on the **5' donor site**, an intronic **branch site**, and the **3' acceptor site**. Alternative splicing allows one gene to produce multiple mRNA/protein isoforms.

Disease anchor	RNA-processing connection in deck
Autosomal-dominant retinitis pigmentosa	Several adRP genes encode pre-mRNA processing/splicing factors, including PRPF-family proteins; photoreceptors have high transcript/protein-processing demands.
Systemic lupus erythematosus	Autoantibodies target nuclear RNA-processing components, including snRNP/Sm-associated antigens discussed in the deck.

SLE framing: the high-yield relationship here is the spliceosomal/snRNP antigen connection, not that every case of SLE is caused by a primary germline spliceosome mutation.

Slides 35-40

Polyadenylation controls stability, translation, and 3' UTR regulation

LO 3

AAUAAA signal transcribed

->

endonucleolytic cleavage ~10-30 nt downstream

->

poly(A) polymerase adds ~80-250 A residues

Poly(A) tail functions: supports transcription termination, nuclear export, mRNA stability, and translation. Progressive tail shortening is associated with reduced translation and eventual mRNA degradation.

Alternative polyadenylation: using different cleavage/poly(A) sites changes 3' UTR length. A shorter 3' UTR may remove miRNA-binding sites and increase protein expression; a longer 3' UTR can preserve more regulatory sites.

Exam traps

Protect polymerase, drug, and processing relationships

All LOs

Trap 1: RNA polymerase does not require a primer.

Trap 2: Bacterial sigma factor selects promoters; it is not the catalytic subunit.

Trap 3: Rifampin targets bacterial RNAP beta; alpha-amanitin targets eukaryotic Pol II.

Trap 4: Pol I makes 45S pre-rRNA; Pol II makes mRNA; Pol III makes tRNA + 5S rRNA.

Trap 5: 5' cap is 7-methylguanosine; the poly(A) signal is AAUAAA near the 3' end.

Trap 6: Alternative splicing changes exon composition; alternative polyadenylation changes the 3' end/UTR.

30-SECOND FINISH

RNA synthesis is 5' to 3' and primer independent. Bacteria: one RNAP + sigma, -10/-35 promoters, rho-dependent/independent termination. Eukaryotes: Pol I rRNA, Pol II mRNA, Pol III tRNA/5S. Rifampin hits bacterial beta; alpha-amanitin hits Pol II. Mature mRNA gets cap + splice + poly(A) tail.